1. Clinical Question & Core Hypothesis

Clinical Question

Pembrolizumab plus platinum chemotherapy should benefit immune-infiltrated (TIL-high) muscle-invasive bladder tumors by combining cytotoxic tumor killing with PD-1 blockade, but some TIL-rich tumors fail to respond. 2) Tier-1 path(s) used: the single path linking pembrolizumab+chemotherapy → cell death and KRAS → cell death: I build on the observation that KRAS signaling modulates cell-death programs and therefore could alter chemotherapy-induced immunogenic cell death and downstream immune-mediated tumor clearance. 3) Tier-2/supporting evidence from the path: chemotherapy (cisplatin) induces cell death and can cause therapy-induced mutations that affect clonal evolution and immunogenicity; these processes could interact with KRAS-driven cell-death modulation to change antigen release or immune visibility. 4) Endpoint choice: I selected therapy_response because the clinical question is whether KRAS status changes sensitivity to the combined pembrolizumab+platinum regimen (most directly assessed by response rates), and because the path emphasizes treatment interactions rather than long-term survival. 5) Deliberate exclusions: I excluded adding other genes or broader mutational burden as anchors to keep the hypothesis mechanistically focused on KRAS as the modulator of therapy-induced immunogenic cell death; other factors (e.g., therapy-induced mutagenesis, cisplatin pharmacodynamics) are listed as secondary modifiers for later exploration?

Working Hypothesis

In muscle-invasive bladder cancer treated with pembrolizumab plus platinum chemotherapy, baseline high tumor-infiltrating lymphocyte (TIL) density predicts greater objective response rate (ORR) overall, but KRAS-mutant tumors within the high-TIL subgroup will exhibit a diminished benefit compared with KRAS-wildtype tumors. This arises because KRAS mutations may modulate chemotherapy-induced immunogenic cell death (ICD) and subsequent T cell–mediated tumor control, reducing the synergistic efficacy of PD-1 blockade and platinum chemotherapy in KRAS-mutant tumors. Therefore, there is a significant interaction between TIL_density (high vs low) and KRAS_mutation status (mutant vs wild-type) on ORR, such that ORR_highTIL_KRASmut < ORR_highTIL_KRASwt, with KRAS status having a lesser or no impact on ORR in TIL_low tumors. Secondary exploratory hypotheses include that therapy-induced mutational processes and cisplatin exposure further modulate this interaction and could influence ICD markers and neoantigen dynamics; these will be evaluated but are not required for the primary mechanistic test.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	KRAS
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

KRAS: KRAS → cell death: I build on the observation that KRAS signaling modulates cell-death programs and therefore could alter chemotherapy-induced immunogenic cell death and downstream immune-mediated tumor clearance.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	therapy_induced_mutations	Supporting modifier from knowledge graph
Secondary	cisplatin	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- therapy_induced_mutations: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- cisplatin: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.